HSU, Ke-Cheng (Clio)

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Summary

PhD candidate in Economics at HKUST, graduating June 2027, using data to inform business decisions. Built structural models of customer behavior, simulated counterfactual policies, and evaluated both profit and customer welfare. Works in Python, R, and C++, uses AI for data management and reporting, and communicates results through clear writing and teaching rated 4.5–4.7 out of 5. Hong Kong permanent resident; native Mandarin, fluent English, communicative Cantonese. Applying for the 2027 Management Trainee Programme, with Customer and Product Development as the preferred function.

Education

The Hong Kong University of Science and Technology, Hong Kong

Doctor of Philosophy in Economics — Expected Graduation: June 2027

Research interests: Empirical industrial organization and digital economics.

Master of Philosophy in Economics — 2020–2022

CGA: 4.1/4.3.

Bachelor of Science in Mathematics and Economics, Minor in Actuarial Mathematics — 2015–2019

CGA: 3.9/4.3. Graduated first in class.

Awards: Dean's List (2016–2019); University Scholarship (730K HKD); Academic Achievement Medal (top 1%); Postgraduate studentship (1.5M HKD); Redbird scholarship (40K HKD).

Experience

Teaching Assistant Coordinator, Department of Economics, HKUST — 2025–2026

- Evaluated teaching assistants' practice teaching, gave feedback on what to change, and helped them improve.

Hall Tutor, UG Hall I, HKUST — 2024–present

- Helped residents work through conflicts and supported student life in a residential community.

Lecturer & Teaching Assistant, Department of Economics, HKUST — 2020–present

- Lecturer for MSc Math Camp (student rating 4.53/5); TA for PhD and undergraduate economics (average 4.72/5 across five courses).

Projects

Return Policy Design in E-Commerce *(Job Market Paper)*

- Built a structural model of purchase, return-shipping-insurance take-up, and return decisions under monopoly pricing, using within-retailer switches on Taobao (2017–2023; six apparel categories; ∼62M RMB revenue; ∼20–30% return rates); scraped and OCR-digitized product attributes in Python.
- Estimated demand, consumer fit uncertainty, and return responses; simulated counterfactual policies and compared seller profit, consumer surplus, and social welfare with bootstrap inference.
- Found that a lenient policy (seller pays return shipping) suits high-margin categories, a strict policy (buyer pays) suits thin-margin ones (improving welfare by ∼50 pp in the thinnest), and a proposed menu (lower price with partial restocking fee vs. full price with full refund) beats both, reaching ∼75% of the best attainable outcome in most categories; back-of-envelope calculation suggests ∼5M RMB profit improvement.

Inventory and Pricing under Resale Price Maintenance (RPM) *(w/ K. Kawaguchi, HKUST; K. Kawai, U. Tokyo)*

- Built a dynamic oligopoly model of pricing and inventory under demand uncertainty and RPM, with equilibrium stockouts and unsold returns (35–40% of Japanese books printed are discarded); constructed panels from monthly bookstore POS data (tens of billions of records across 1,000+ stores), linked via ISBNs and merged with demographics and title-level information scraped from Amazon.
- Estimated demand and supply with instrumental variables; validated the estimator with Monte Carlo; implemented equilibrium simulation in C++.
- Found that the static model overestimates marginal cost by ∼20% when ignoring dynamic demand uncertainty and inventory; uncovered stockout-driven substitution patterns relevant for inventory and assortment decisions.

Skills

Languages: Mandarin (Native), English (Fluent), Cantonese (Communicative).
Technical: Python, R, C++, Stata, LaTeX, AWS, HPC/SLURM; structural estimation and counterfactual simulation.
AI: Uses AI tools for data management, coding, and generating weekly reports.
References: Kei Kawai (UTokyo; Compass Lexecon), Kohei Kawaguchi (HKUST; UTokyo Econ Consulting), Pak Hung Au (HKUST).